



Play with AI (PL-AI): A play-centered, design-based curriculum for AI literacy in pre-K and kindergarten

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ABSTRACT

This paper reports on the design, implementation, and refinement of Play with AI (PL-AI), a developmentally appropriate curriculum designed to introduce foundational artificial intelligence (AI) concepts to pre-kindergarten and kindergarten children. Using a design-based research (DBR) approach, two pre-K and two kindergarten teachers co-designed, piloted, and refined seven sequenced activities integrating unplugged play (e.g., children composing and testing “how-to” algorithms), tangible coding (e.g., programming Bee-Bot or Ozobot to follow paths), and guided dialogue with a social AI robot (e.g., comparing children's story endings with the robot's). The study addressed two research questions: (1) How does teacher participation in co-design shape implementation feasibility, pedagogical confidence, and curriculum ownership during the enactment of a play-based AI literacy curriculum? and (2) What design principles emerge as feasible and sustainable for embedding AI literacy in early childhood classrooms? Data sources included teacher surveys, 32 h of classroom video, researcher field notes, and design-meeting transcripts. Findings are presented as qualitative and formative evidence documenting patterns of enactment, teacher adaptation, and children's emerging reasoning. Quantitative tallies and observational rubrics are reported as descriptive indicators to inform iterative refinement, not as validated outcome measures or statistically generalizable learning gains. Teachers' confidence and pedagogical agency increased substantially, with co-design participation fostering curriculum ownership, reflective practice, and meaningful adaptation. Across iterative cycles, four interrelated design principles emerged—embodied play, tangible coding, guided dialogue, and teacher co-design—supporting conceptual clarity, developmental appropriateness, and sustainability. This study contributes to the emerging field of early AI education by offering empirically grounded design knowledge aligned with the AI4K12 Initiative (2024) and NAEYC (2021) frameworks, and provides practical guidance for integrating AI literacy through play-based, ethical, and collaborative pedagogy.

1. Introduction

Artificial intelligence (AI) increasingly shapes how people work, learn, and interact—from navigation and content curation to conversational assistance. Children encounter AI both explicitly (e.g., through voice assistants, adaptive games, and social robots) and implicitly (e.g., via recommendation engines and predictive systems that mediate media use and information access). Despite this pervasive presence, the pedagogical implications of AI in early childhood remain underexplored (Lee, 2026a, 2026b; Lee, Lee, & Yoon, 2025), as most educational policies and curricula continue to prioritize older learners. This imbalance underscores the need for developmentally appropriate approaches to foundational AI concepts—such as algorithms, data, and fairness—while carefully attending to ethical responsibilities related to young children's participation and well-being.

To address this gap, the Play with AI (PL-AI) curriculum was developed to ground AI ideas in concrete, familiar classroom experiences. Instead of introducing AI as an abstract technological domain, PL-AI situates learning within embodied, play-based contexts that align with early childhood pedagogy. Through hands-on activities, children

engaged in playful tasks such as giving step-by-step instructions to a floor robot, debugging navigation paths during block play, and comparing their own story endings with those generated by a conversational AI robot. These experiences positioned AI as a human-designed, instruction-following system, countering interpretations of AI as autonomous or magical.

Recognizing that children's talk about animacy and agency reflects both developmental reasoning and broader societal narratives about AI, PL-AI frames questions (e.g., whether robots have feelings) as inquiry-based explorations rather than facts to memorize (Lee, Jo, et al., 2025). In doing so, the curriculum integrates conceptual understanding with ethical reflection, supporting young children in developing both analytical insight and critical awareness. By embedding AI concepts within movement, storytelling, and collaborative problem-solving, PL-AI aligns with developmentally appropriate practice (NAEYC, 2021) while fostering early ethical engagement with emerging technologies.

These examples are intentionally early childhood specific: they rely on short, embodied action sequences, concrete manipulatives, small-group talk, and familiar routines (centers, block play, story time) so

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that abstract AI ideas are experienced through guided play instead of being explained through technical definitions (Lee & Kim, 2025). Throughout the paper, PL-AI is positioned not as a general K–12 framework, but as an early childhood translation of AI4K12's concepts aligned with NAEYC's developmentally appropriate practice.

The AI4K12 Initiative identifies “Learning from Data” and “Societal Impact” as two of its “Five Big Ideas,” thereby linking computational and ethical understanding in early learning contexts (Touretzky et al., 2019). Recent scholarship has demonstrated that young children can engage with algorithmic reasoning through sequencing and patterning (Bers, 2020; Lee et al., 2022; Strawhacker & Bers, 2019) and begin to recognize the social dimensions of data and fairness through guided exploration of bias and inclusion (Druga et al., 2019; Kizilcec, 2022). Together, this scholarship suggests that foundational AI ideas can be meaningfully introduced in play-based environments when carefully translated for developmental appropriateness.

AI literacy is increasingly conceptualized as a multidimensional construct encompassing understanding what AI is and how it functions, evaluating its strengths and limitations, and using it responsibly (Ng et al., 2021; Touretzky et al., 2019). The AI4K12's guidelines (AI4K12 Initiative, 2024) further organize this knowledge into five “Big Ideas”: Perception, Representation and Reasoning, Learning, Natural Interaction, and Societal Impact. Recent guidance from organizations such as Digital Promise (2024) and Hibbert et al. (2024) highlights the importance of moving beyond operational familiarity with AI tools toward fostering critical and ethical engagement. However, these frameworks were largely developed for older students and therefore require adaptation for early childhood settings, where learning is best supported through play, storytelling, collaboration, and hands-on exploration aligned with developmentally appropriate practices (Lee & Kim, 2025; NAEYC, 2021).

Parallel work in computational thinking (CT) and educational robotics provides a foundational bridge. Research has shown that coding and tangible robots such as Bee-Bot, KIBO, and Ozobot promote sequencing, cause–effect reasoning, and debugging, while also supporting persistence, collaboration, and problem-solving (Bers, 2018; Lee, Jo, et al., 2025; Su & Yang, 2023; Sullivan & Bers, 2019). Meta-analyses have reported gains in both algorithmic reasoning and socio-emotional skills (Hong, 2024; Su & Yang, 2023). Emerging studies in early AI education similarly demonstrate that young children can engage with concepts such as training data, fairness, and human–AI interaction when supported through play-based activities (Chen et al., 2023; Druga et al., 2019; Hitron et al., 2019; Lee, Jo, et al., 2025; Strawhacker & Bers, 2015). Yet few interventions explicitly link CT experiences to AI concepts such as data-driven decision-making, human design, or distinctions between living and non-living systems—raising concerns that children may perceive “smart” devices as magical rather than human-designed and instruction-following systems.

This study addresses these gaps by reporting on the design process and analysis of Play with AI (PL-AI), a play-based AI curriculum for pre-kindergarten and kindergarten. Guided by a design-based research (DBR) approach, the study investigates two research questions: (1) How does teacher co-design influence implementation quality, confidence, and ownership, as reflected in self-reports, adaptation of activities, and willingness to share practices with peers?; and (2) What design principles emerge as feasible and sustainable for embedding AI literacy in early childhood classrooms?

2. Conceptualizing AI literacy for early childhood

To address the number of concept lists referenced across the manuscript, PL-AI's conceptual foundations are organized into three recurring concept categories (a) AI literacy concepts (e.g., data, human design, limits, fairness), (b) computational: thinking concepts (e.g., sequencing, debugging, pattern recognition, abstraction), and (c) ethical/social concepts (e.g., bias, accountability, inclusion, responsible use). This

categorization makes explicit how PL-AI integrates AI4K12 “Big Ideas” with developmentally appropriate pedagogies and provides a consistent lens for describing activities, teacher prompts, and analytic codes.

Building on these categories, frameworks such as AI4K12 (AI4K12 Initiative, 2024) articulate five “Big Ideas” aimed at specifying what K–12 learners should know about AI. More recent efforts emphasize criticality and responsibility in everyday use (Digital Promise, 2024) and propose a developmental progression from understanding to applying, evaluating, and ultimately creating AI systems (Hibbert et al., 2024). Chiu (2024) further clarifies distinctions between AI literacy (knowledge and awareness) and AI competency (effective application), underscoring the need for instruction that cultivates both. For young children, AI literacy must be translated into developmentally appropriate experiences—hands-on, playful, social, language-rich, and culturally responsive (NAEYC, 2021).

Accordingly, an early-childhood AI curriculum should: (a) reveal the human-designed, instruction-following nature of “smart” systems; (b) connect inputs (data) to outputs (behavior); and (c) invite dialogue about fairness, bias, and limits—using familiar contexts and accessible language. Table 1 synthesizes these dimensions and sources into a practical scaffold for the PL-AI design.

In summary the PL-AI framework synthesizes seven mutually reinforcing dimensions of AI literacy for early childhood. First, understanding/knowing establishes foundational knowledge about what AI is and how it works—basic ideas like algorithms, data, and bias—so children begin to see “smart” behavior as rule-based rather than magical (Digital Promise, 2024; Ng et al., 2021). Building on this, application/use emphasizes engaging with age-appropriate AI tools to solve problems or create content (e.g., coding a robot to follow a path or prompting a storytelling agent), translating concepts into purposeful action (Hibbert et al., 2024; Lee, 2019; Ng et al., 2021). Evaluation/critical thinking invites learners to question AI outputs—Is it accurate? Could it be biased? How transparent is the process?—laying early habits of evidence-seeking and skepticism (Chiu, 2024; Ng et al., 2021). Integrated throughout is ethics and societal impact, which frames discussions about fairness, accountability, and safety in concrete, child-meaningful scenarios (e.g., “Was that choice fair?” “Who decided what the robot knows?”), making visible the human decisions behind AI (Long & Magerko, 2020; Touretzky et al., 2019). Communication and collaboration focus on explaining how AI behaves and working with peers to interpret or design systems, aligning with social, language-rich learning in early grades (Digital Promise, 2024).

Creation/design moves children from users to novice designers—co-

Table 1
Dimensions of AI literacy and supporting references.

Dimension	Description	Example Sources
Understanding/ Knowing	Foundational knowledge of what AI is, how it works (algorithms, data, bias).	Ng et al., 2021; Digital Promise, 2024
Application/Use	Ability to engage with AI systems to solve problems or create content.	Ng et al., 2021; Hibbert et al., 2024
Evaluation/Critical Thinking	Assessing accuracy, bias, transparency, and reliability of AI outputs.	Ng et al., 2021; Chiu, 2024
Ethics & Societal Impact	Awareness of fairness, accountability, safety, and broader social implications.	Bers, 2022; Long & Magerko, 2020; Touretzky et al., 2019
Communication & Collaboration	Explaining AI behavior and working with others to interpret or design AI.	Digital Promise, 2024
Creation/Design	Developing or co-designing AI systems (e.g., data choices, prompt design).	Hibbert et al. (2024)
Developmentally Appropriate Practice	Adapting AI ideas into play-based, social, and exploratory learning.	Bers, 2018; Lee, Jo, et al., 2025; NAEYC, 2021

authoring data choices or prompts and iteratively refining instructions—thereby strengthening agency and metacognition (Hibbert et al., 2024; Ng et al., 2021). Finally, developmentally appropriate practice anchors all dimensions in play-based, hands-on, culturally responsive activity structures—roleplay, manipulatives, and guided dialogue—so abstract AI ideas are experienced through exploration and talk rather than lecture (Bers, 2018; Lee & Kim, 2025; NAEYC, 2021). While AI literacy frameworks articulate what learners should understand, computational thinking research offers insight into how young children can develop the reasoning skills that make such understanding possible.

2.1. Computational thinking and educational robotics as foundations for AI education

Research on computational thinking (CT) in early childhood provides a solid pedagogical base for introducing artificial intelligence (AI). CT emphasizes problem decomposition, pattern recognition, abstraction, sequencing, and algorithmic reasoning—skills that map closely onto how AI systems are built and understood (Grover & Pea, 2018; Lee, 2026a, 2026b; Wing, 2017). In early-learning “playgrounds,” children engage with these concepts by sequencing steps, exploring cause–effect relations, and debugging errors through block-based programming environments and tangible robots (Bers, 2018; Lee, 2019; Sullivan & Bers, 2019). Studies with platforms such as ScratchJr, KIBO, and Bee-Bot demonstrate that preschool and kindergarten learners can construct simple algorithms, test predictions, and revise procedures when supported by guided play (Strawhacker & Bers, 2015; Kazakoff & Bers, 2014).

Systematic reviews and meta-analyses validate these findings, reporting consistent gains in algorithmic reasoning, spatial and logical thinking, and collaborative persistence after coding or robotics interventions (Ching & Hsu, 2023; Hong, 2024; Su & Yang, 2023). Importantly, CT experiences also foster learning dispositions—perseverance, curiosity, and cooperative problem solving—that mirror the habits of mind required to reason about AI systems (Bers, 2022). Robotics activities invite children to give precise instructions, observe immediate feedback, and engage in group debugging, all of which make abstract computational principles visible and negotiable through dialogue (Bers, 2018; Sullivan & Bers, 2019).

Yet, despite this promising evidence base, much of the CT literature treats programming as an end in itself or as a pathway to later computer science, rather than as an explicit bridge to AI concepts. Few empirical studies examine how early coding practices can illuminate data-driven prediction, model limitations, or ethical considerations in machine behavior (Cukurova et al., 2024). Research involving robots or conversational agents in early childhood education settings often highlights usability or motivation but rarely addresses how children conceptualize animacy, agency, and the human design behind algorithms (Lee, Jo, et al., 2025; Zeng et al., 2023). Without targeted scaffolding, young learners may misinterpret “smart” devices as autonomous or sentient—believing, for instance, that a robot “chooses” or “feels” rather than following instructions encoded by people (Druga et al., 2019).

Addressing this instructional gap requires curricula that position CT not only as a technical skillset but also as a gateway to understanding AI’s data-driven and socially constructed nature. By coupling block-based coding and robotics with explicit discussion of datasets, bias, and the distinction between living and non-living systems, educators can help children see that AI operates through human-curated rules and information rather than independent will. This integration situates CT as a developmentally appropriate pathway to AI literacy, preparing young learners to explore how algorithms and data underpin intelligent behavior while maintaining the playful, inquiry-oriented spirit essential in early-childhood classrooms (Lee & Yoon, 2008). Beyond cognitive and procedural skills, AI literacy also requires attention to ethical and societal dimensions.

2.2. Ethical and societal dimensions in early AI learning

Ethical and societal considerations are foundational components of AI literacy (Long & Magerko, 2020; Touretzky et al., 2019). Frameworks such as AI4K12 (AI4K12 Initiative, 2024) and Digital Promise (2024) explicitly include fairness, accountability, and the social impact of intelligent systems as core learning goals, underscoring that understanding how AI works must be coupled with reflecting on how it should be used. AI4K12’s “Five Big Ideas” (e.g., AI as perception, representation, learning, natural interaction, and societal impact) and Digital Promise’s dimensions of AI literacy (e.g., understanding, application, evaluation, ethics, and collaboration) overlap in highlighting not only technical knowledge but also ethical reflection. This perspective aligns with work emphasizing that programming and AI activities can function as sites for learning human values—such as fairness, responsibility, and care—rather than merely technical skills (Bers, 2022).

For young children, these concepts are translated into developmentally appropriate practices that render ethical questions visible within everyday contexts. Early childhood classrooms, in particular, constitute rich sites for value-oriented dialogue when ethical concepts are intentionally embedded in stories, classroom routines, and collaborative play. For example, when a robot repeatedly selects the same ending or responds unevenly to prompts, educators can guide learners to ask, “Was that fair?” or “Who decided what the robot knows?”—questions that help children connect algorithmic behavior to human choices and data sources. Research shows that such discussions demystify AI, counteract misconceptions about animacy, and foster a sense of agency and critical stance (Bers, 2018; Zeng et al., 2023). Integrating these inquiries with NAEYC’s (2021) developmentally appropriate practice ensures conversations remain curiosity-driven, collaborative, and culturally responsive, allowing children to link emerging AI concepts with the ethical and social frameworks shaping their everyday lives.

2.3. Teacher agency, professional learning, and sustainable innovation

Teacher agency—the capacity to adapt, evaluate, and take ownership of new practices—is a decisive factor in whether technological innovations are meaningfully implemented (Penuel et al., 2015; Priestley et al., 2015; Voogt et al., 2015). In early childhood settings, this agency is shaped by the complex ecology of routines, relationships, and resource constraints. Research shows that teachers who feel confident and supported are more likely to experiment with emerging tools, integrate them into centers, and sustain their use beyond initial trials (McKenney & Reeves, 2019). Professional learning that is hands-on, collaborative, and ongoing strengthens teachers’ content knowledge, pedagogical repertoire, and sense of efficacy—key elements for high-fidelity implementation of technology innovations (Cukurova et al., 2024).

Design-based research (DBR) provides a methodological architecture for cultivating this agency. Through iterative cycles of co-design, enactment, and reflection, teachers and researchers collaboratively shape materials, examine how activities fit within everyday practice, and refine designs in response to authentic challenges (Brown, 1992; Design-Based Research Collective, 2003; Wang & Hannafin, 2005). DBR’s flexibility and explicit attention to theoretical and practical rationales are particularly valuable in early-childhood classrooms, where pacing, space, and instructional routines differ across contexts.

For emerging areas such as AI literacy, co-design enables teachers to contribute to both the conceptual alignment and the practical feasibility of interventions. Educators help shape the language used to explain abstract concepts, select props or story contexts that resonate with young learners, and adjust schedules to embed activities in familiar centers (e.g., block play, dramatic play, or story time). Such participation not only enhances the local relevance of curricula but also supports teachers’ identity as knowledgeable innovators who can guide children’s reasoning about fairness, bias, and the human role in technology design (Bers, 2018; Sullivan & Bers, 2019). Studies of teacher professional

development in STEM and coding similarly show that collaborative design experiences promote deeper pedagogical reasoning, richer facilitation of inquiry, and more equitable classroom dialogue (Hong, 2024; Su & Yang, 2023). By embedding AI literacy within sustained, participatory professional learning, projects like PL-AI foster not only immediate instructional capacity but also the long-term sustainability needed to normalize ethical, developmentally appropriate AI education in the early years.

2.4. The state of early childhood AI interventions: What's missing

Despite a surge of interest in AI education, research aimed at preschool and kindergarten learners remains surprisingly limited. Systematic reviews indicate that most AI-literacy interventions focus on upper elementary or secondary students, emphasizing tool use or basic coding skills rather than cultivating deeper conceptual understanding (Chiu, 2024; Su & Yang, 2022; Zeng et al., 2023). When robots or conversational agents are introduced in early-childhood classrooms, studies tend to highlight engagement—such as attention, excitement, or social presence—without systematically investigating how young children reason about algorithms, data, or the human design that underpins intelligent behavior (Druga et al., 2019; Su & Yang, 2023). Only a small number of empirical efforts deliberately connect the AI4K12's “Big Ideas” or ethical reasoning to pedagogy for children under six. Long and Magerko (2020), for example, argue that datasets and algorithmic rules should be discussed even with novices, yet their work offers limited classroom evidence with preschool populations. Similarly, research on educational robotics occasionally notes how children attribute feelings or intentions to machines, but rarely incorporates structured conversations about fairness, bias, or accountability alongside programming tasks (Bers, 2022; Sullivan & Bers, 2019).

Teacher preparation remains another underdeveloped dimension. Reviews show that professional learning for early-childhood educators seldom addresses AI content knowledge or strategies for scaffolding ethical dialogue (Cukurova et al., 2024), leaving practitioners with little guidance for moving beyond surface-level enthusiasm toward rich discussions about human-machine boundaries, data provenance, and design choices. Collectively, the literature suggests persistent gaps in early-childhood AI education: few curricula translate AI literacy into hands-on, play-based sequences appropriate for preschoolers; ethical and societal perspectives are often fragmented or absent rather than woven into everyday routines; teachers are rarely positioned as co-designers, which limits ownership and contextual fit; and there is little evidence on formative assessment approaches that capture children's emerging understanding of AI while preserving the playful character of early learning environments.

2.5. Contribution of PL-AI study

This study addresses these gaps by reporting an empirically developed, play-based AI curriculum for pre-K and kindergarten (Play with AI, PL-AI). Unlike prior interventions that introduced isolated AI concepts through short-term activities (e.g., Druga et al., 2019; Hitron et al., 2019; Strawhacker & Bers, 2015), PL-AI integrates a sequenced set of seven activities that progress from unplugged play to tangible coding to social-robot dialogue. The curriculum deliberately embeds ethical reasoning and animacy discussions alongside algorithmic practice, making fairness, bias, and human design visible in everyday routines. A further contribution is the use of lightweight, observation-friendly assessments (rubrics, debugging counts, concept-statement tallies) that capture learning without disrupting play, offering a replicable model for busy early-childhood classrooms. Together, these features differentiate PL-AI as both a curricular and methodological innovation in early AI education.

3. Methods

This study employed a design-based research (DBR) approach to create, implement, and refine the Play with AI (PL-AI) curriculum in collaboration with early childhood teachers. DBR was selected because it enables the concurrent development of practical innovations and theoretical insights while remaining responsive to classroom complexity (Brown, 1992; Design-Based Research Collective, 2003). Iterative cycles of design, enactment, analysis, and revision allowed the researcher and teachers to adapt activities to children's responses and to the constraints of authentic learning environments, generating evidence about both feasibility and learning processes.

Consistent with the epistemological commitments of design-based research, the purpose of this study was not to test hypotheses, estimate effect sizes, or establish causal learning outcomes. Instead, data were used formatively to examine implementation feasibility, patterns of enactment, teacher sensemaking, and emergent learning processes as the PL-AI curriculum evolved through iterative cycles of co-design, implementation, and refinement (Brown, 1992; McKenney & Reeves, 2019). Quantitative tallies and observational rubrics served descriptive and analytic functions within the design process and are not interpreted as validated outcome measures.

3.1. Ethical consideration and consent

This study was reviewed and approved by the university's Institutional Review Board (IRB) prior to data collection. Written informed consent was obtained from all participating teachers and from parents or legal guardians of all child participants. Children provided age-appropriate verbal assent before participating in activities and data collection. All procedures adhered to ethical guidelines for research with young children, including protections for confidentiality, voluntary participation, and the right to withdraw at any time without penalty. Video recordings and observational data were securely stored, de-identified for analysis, and used solely for research purposes.

3.2. Research design

This single-case DBR project proceeded through four phases—(1) teacher recruitment and training, (2) co-designing the curriculum, (3) implementation, and (4) refinement—each generating artifacts, reflections, and data feeding subsequent iterations. The study adopted a conjecture-mapping stance (Sandoval, 2014), linking activity features to mediating processes and intended outcomes (e.g., role-play → algorithm talk; social-robot storytelling → discussions of feelings, fairness, and limits). Table 2 presents the DBR cycle and associated activities.

3.3. Context and participants

This foundation-funded study was conducted with four early

Table 2
Four phases of developing the PL-AI curriculum.

Phase	Focus	Activities
Phase 1	Teacher Recruitment & Interactive Training	Conducting pre-survey, three meetings introducing the PL-AI project, DBR principles, AI literacy, and developmentally appropriate pedagogy
Phase 2	Co-Designing the Curriculum	Two workshops using codable and AI-enabled robots; brainstorming and prototyping activities
Phase 3	Implementation	One month of PL-AI stations during free-play (20–40 min); weekly researcher-teacher check-ins
Phase 4	Refinement	Final meeting to share feedback, align with guiding frameworks, revise activities, and conducting a post-survey

childhood educators—two teaching pre-kindergarten and two teaching kindergarten—in an urban district located in the southern United States. Both participating schools were Title I campuses serving children from families with low socioeconomic backgrounds. The participating teachers represented diverse backgrounds and teaching experience levels. One was a White kindergarten teacher in her twenties with two years of experience teaching kindergarten and second grade. Another was a White pre-kindergarten teacher in her forties who had more than twenty years of experience across grades pre-K through fifth. A third participant was a Latina pre-kindergarten teacher in her forties with approximately fifteen years of teaching experience in early elementary grades. The fourth was a Latina kindergarten teacher in her fifties with over two decades of experience working with children in kindergarten through second grade.

3.4. Training, Co-design, and implementation process

Development of the Play with AI (PL-AI) curriculum unfolded through four iterative phases typical of design-based research. Following recruitment, four teachers participated in interactive training sessions that introduced the project goals, DBR principles, co-designing, foundational AI concepts, and developmentally appropriate pedagogy. These sessions also helped teachers become familiar with the guiding frameworks (AI4K12 and NAEYC) and explore the basic functions of the robots and materials to be used with children. At the same time, this promoted joint engagement through knowledge sharing (Penuel et al., 2022). During this session, the goals of AI education in early childhood (KinderLab Robotics, 2023) were introduced, and teachers discussed how these objectives aligned with their own instructional priorities. They also examined the connections between the AI4K12 “Big Ideas” and NAEYC’s principles of developmentally appropriate practice (DAP). Table 3 presents the AI education goals embedded in the PL-AI curriculum prototype aligned with AI4K12 and NAEYC’s DAP principle, and Table 4 presents the AI literacy components embedded in the PL-AI curriculum aligned with AI4K12 and NAEYC’s DAP principles.

The teachers also reviewed literature on AI literacy and aligned its key components with the AI4K12’s “Big Ideas” and NAEYC’s principles of DAP, which together served as the foundational framework for the PL-AI curriculum prototype.

Building on this foundation, teachers and the researcher engaged in co-design workshops where they engaged in various discussions, planning lessons, and exploring codable and AI-enabled robots (e.g., Bee-Bot

Table 3

AI education goals in the PL-AI curriculum prototype.

Code	AI Education Goal	Aligned AI4K12 Big Ideas	NAEYC’s DAP Connection
1	AI is Created by People – Children recognize AI as designed by humans.	Representation & Reasoning; Societal Impact	Builds curiosity, agency, and awareness of human roles in innovation
2	AI Follows Instructions to Work – AI works by following programmed rules.	Representation & Reasoning; Learning	Encourages step-by-step exploration, persistence, and guided practice, supporting problem-solving and resilience in play
3	AI and Living Things —Children explore differences between living things and AI/robots (e.g., needs, growth, feelings) and discuss what robots can/cannot do.	Societal Impact	Supports science inquiry and discussion about living vs. non-living
4	AI Helps People – AI supports but does not replace human decision-making.	Natural Interaction; Societal Impact	Promotes collaborative, equitable, and real-world connections

Table 4

AI literacy components in the PL-AI curriculum prototype.

Code	AI Literacy Component	Aligned AI4K12’s Big Ideas	NAEYC’s DAP Connection
1	Understanding algorithms and step-by-step processes	Representation & Reasoning	Supports sequencing, routines, and problem-solving through hands-on play
2	Programming and coding (sequencing commands, directionality, loops)	Representation & Reasoning; Learning	Encourages active exploration and iterative learning through trial and error
3	Debugging and evaluation (fixing mistakes, revising steps, testing outcomes)	Learning	Builds persistence, resilience, and reflection—key to play-based problem-solving
4	Critical thinking and social interaction (ethics, collaboration, comparing human vs. AI thinking)	Societal Impact; Natural Interaction	Promotes collaboration, communication, equity, and ethical reasoning in context of technology

[button-based floor robot], Ozobot [line-following, color-code robot], Rivgot [barcode-scanning drawing robot], and Miko [social AI robot]) and brainstormed how to integrate AI concepts—such as algorithms, debugging, fairness, and animacy—into familiar centers and story routines.

During these sessions, they drafted provisional activities to be piloted with children. Early ideas included: “*Move Like a Robot*,” an unplugged game where children created and followed step-by-step instructions for physical movements, highlighting that “smart” systems require explicit directions. “*Story Sequencer*,” in which groups arranged picture cards to illustrate a narrative, then “programmed” a classmate to act out the story, surfacing the importance of order and clarity. “*Data Detective*,” where children collected simple data about favorite colors or animals, graphed the results, and discussed how information can guide machine decisions. “*Robot Helpers*,” a dramatic-play center in which children role-played service robots and “engineers,” deciding how machines should help people. “*Sensor Safari*,” using classroom objects to demonstrate how robots detect signals (e.g., light or color), sparking talk about perception and limits.

From these provisional designs, the team built a structured sequence of seven pilot activities—*Program “My Moves,” “How-To Algorithms,” “Treasure Code Quest,” “Code the Path,” “Color Code Stories,” “Scan and Draw,”* and *“AI Stories and Me”*—each aligning an AI literacy component with a child-facing goal. Teachers tested these prototypes during a five-week implementation period. Sessions were scheduled two or three times per week (20–40 min), rotating so small groups of two to five children could participate while teachers facilitated or observed. Materials included instruction cards, laminated mats, Bee-Bot, Ozobot, Rivgot (a barcode-scanning robot for programmable actions), and Miko. Classroom implementation generated refinements to the original plans. For example, “*Program My Moves*” evolved to include peer “debugging” rounds, while “*How-To Algorithms*” incorporated a reflection step where children checked if their directions were “clear enough for a robot.” Teachers modified “*Treasure Code Quest*” by adding number or letter clues, linking navigation with literacy and numeracy skills. “*Color Code Stories*” expanded to include cooperative storytelling, and *Scan and Draw* gained a prediction element where children guessed what image the robot would retrieve. “*AI Stories and Me*” developed prompts comparing the robot’s story endings with children’s own, prompting ethical discussions about bias and creativity. Table 5 presents the activities, AI connections, and expected child outcomes in the PL-AI curriculum.

Throughout implementation, teachers adapted prompts, props, and scheduling to suit classroom rhythms—for instance, embedding algorithm tasks in literacy corners or adding debugging challenges to block play. Weekly virtual check-ins enabled teachers and the researcher to share observations, troubleshoot, and adjust activities. A final reflection

Table 5

PL-AI curriculum: Activities, AI connections, and child outcomes.

Activity (New Name)	Description (AI Connection)	AI Literacy Component(s)	AI Education Goal(s)	Objectives (Child Outcomes)	Assessment(s)
1. Program My Moves (Unplugged – Role Play)	Children act as “robots,” following peer/programmer commands; demonstrates that AI follows rules created by people.	1	1, 2, 3	Follow & give clear commands; understand AI doesn't act independently.	Accuracy of commands; cause–effect reasoning.
2. How-To Algorithms (Unplugged)	Children draft or draw step-by-step instructions for simple tasks; teacher enacts them exactly, prompting debugging.	1, 2, 3, 4	1, 2, 4	Create/revise algorithms; understand AI processes directions.	Logical order; clarity; debugging ability.
3. Treasure Code Quest (Unplugged/Plugged)	Children map paths or code a digital character, illustrating that AI navigation is human controlled.	1, 2	1, 2, 3	Design and follow codes; link commands to outcomes.	Navigation accuracy; peer collaboration.
4. Code the Path (Plugged/Bee-Bot)	Children program a robot to complete math or literacy tasks; teachers highlight its chip/battery vs. a brain.	2, 3	1, 2, 3, 4	Program & debug robots; discuss human design in tech.	Algorithm accuracy; debugging; turn-taking; collaboration.
5. Color Code Stories (Plugged/Ozobot)	Children design color paths that the robot “reads,” showing how AI interprets human signals.	1, 2	1, 2, 4	Create/test color-coded algorithms; see AI interpret data.	Correct coding; problem-solving.
6. Scan and Draw (Plugged/Rivgot)	Children scan barcodes linked to words or drawings, learning how AI reads encoded data.	1, 3	1, 2, 3, 4	Connect inputs to outputs; see AI as helper, not thinker.	Accuracy of scans; troubleshooting.
7. AI Stories and Me (Plugged/Humanoid Robot-Miko)	Children compare human and AI storytelling; discuss AI's limits and purpose.	1, 2, 3, 4	1, 2, 3, 4	Contrast AI vs. human stories; reason ethically about tech.	Contrast AI vs. human stories; reason ethically about tech.

meeting allowed participants to review video excerpts, evaluate which provisional ideas became robust, and finalize the refined activity set, which is summarized later.

4. Results

Research Question 1. How does teacher participation in the co-design process shape implementation feasibility, pedagogical confidence, and curriculum ownership during the enactment of a play-based AI literacy curriculum in pre-kindergarten and kindergarten classrooms?

4.1. Teacher agency, confidence, and curriculum ownership

Survey results and meeting transcripts indicated a notable increase in teachers' confidence and sense of agency throughout the study. Rather than being interpreted as statistically generalizable gains, changes in self-ratings are reported as descriptive indicators of shifting teacher perceptions across the design cycles. Average self-ratings on a 5-point Likert scale measuring comfort with teaching AI-related topics increased across phases, reflecting growing familiarity and confidence as teachers engaged in co-design, enactment, and reflection.

Teachers consistently attributed this shift to their active participation in the co-design process, which enabled them to align AI concepts with classroom routines, instructional standards, and available resources. During post-implementation meetings, teachers described how co-design workshops and hands-on robot trials helped them visualize integration into daily centers. One teacher explained that because the activities were built collaboratively, she felt confident embedding them into literacy and math centers rather than treating them as stand-alone lessons.

These accounts are presented as qualitative evidence of teacher sensemaking and professional growth, rather than as measures of instructional effectiveness. Teachers reported increased willingness to experiment, adapt prompts, and troubleshoot alongside children, indicating reduced anxiety around AI-related technologies.

Teachers also demonstrated curriculum ownership by modifying materials and integrating AI activities with core content. For example, one teacher adapted the Scan and Draw station by inserting laminated vocabulary cards so that the robot could retrieve and draw familiar words. Others mapped Bee-Bot routes onto number grids or alphabet mats, integrating AI exploration with numeracy and literacy goals. Such

adaptations are reported as observed patterns of enactment that illustrate feasibility and contextual fit, rather than standardized outcomes.

Several participants extended their engagement beyond the study by informally sharing PL-AI activities with colleagues during planning meetings or after school. These instances are interpreted as indicators of perceived practicality and sustainability of the curriculum, not as evidence of diffusion or impact at scale.

4.2. Refining activities through iteration

The implementation phase revealed which activities were most workable within classroom routines and which required refinement. Several provisional activities developed during co-design were streamlined or merged to improve clarity and engagement. *Move Like a Robot* was combined with *Program My Moves* to create a more coherent entry point for algorithmic thinking, while *Story Sequencer* was condensed into *How-To Algorithms* to maintain momentum while preserving sequencing goals.

Two early prototypes—*Data Detective* and *Sensor Safari*—were discontinued due to logistical constraints. These decisions were informed by formative observations indicating that data collection tasks diverted attention from conceptual reasoning and that sensor calibration challenges disrupted play flow. Key ideas from both activities were absorbed into *Scan and Draw* and *Color Code Stories*, where relationships between inputs and robot responses were more immediately visible to children.

Teachers also formalized spontaneous behaviors observed during implementation. For example, children's informal “races” to complete robot paths were redesigned as structured challenges emphasizing collaboration, turn-taking, and debugging. These refinements illustrate how classroom enactment informed design decisions across cycles, a central feature of design-based research.

4.3. Synthesis of results

In synthesis, the results document patterns of teacher engagement, curricular adaptation, and classroom enactment that informed the refinement of the PL-AI curriculum. Without making claims of causal effectiveness, the findings illustrate how teacher co-design supported implementation feasibility, increased pedagogical confidence, and fostered curriculum ownership. Across classrooms, children's participation was characterized by sustained engagement, frequent peer collaboration, and emerging use of language related to sequencing,

instructions, and fairness. These observations are reported as formative indicators of emerging reasoning within play-based contexts, not as validated measures of learning outcomes.

The results therefore address [Research Question 1](#) by showing how co-design functioned as a mechanism for strengthening implementation quality and sustainability, and they provide the empirical grounding for the design principles articulated in response to [Research Question 2](#).

4.4. Transition to curriculum design principles

Insights from teacher participation extended beyond individual confidence and ownership to shape the evolving design of the PL-AI curriculum itself. As teachers adapted, combined, and refined activities, their decisions illuminated which pedagogical structures were most feasible and sustainable in early learning contexts. These iterative modifications—guided by classroom realities, children's engagement, and ethical dialogue—contributed to the identification of key design principles that anchored the final version of PL-AI.

The following section synthesizes findings related to [Research Question 2](#), detailing the emergent design principles that characterize effective and enduring integration of AI literacy in early childhood classrooms.

Research Question 2. What design principles emerge as feasible and sustainable for embedding AI literacy in early childhood classrooms?

4.5. Synthesis of learning outcomes

Building upon the teacher-driven refinements identified through co-design, analysis of classroom observations and reflective discussions revealed how these adaptations translated into children's learning outcomes and informed the articulation of sustainable design principles. The PL-AI curriculum supported three interrelated strands of development—conceptual, procedural, and dispositional—consistent with frameworks describing learning as the integration of knowledge, practice, and disposition (Bers, 2022; Brennan & Resnick, 2012; NAEYC, 2021).

These strands interacted dynamically: conceptual understanding of AI as human-created and rule-based emerged through embodied play; procedural fluency in sequencing and debugging developed through tangible coding activities; and dispositional growth in persistence, curiosity, and collaboration was fostered by a supportive, play-rich environment. Ethical reflection—particularly around fairness and human decision-making—surfaced naturally through guided dialogue, linking children's reasoning about AI behavior to broader social and moral considerations.

This multidimensional learning progression not only demonstrated the curriculum's developmental appropriateness but also revealed the underlying pedagogical mechanisms that made the PL-AI model effective and replicable. These mechanisms were later synthesized into a set of emergent design principles, which articulate how embodied play, tangible coding, guided dialogue, and teacher co-design collectively sustain meaningful and equitable early AI education.

4.6. Emergent design principles

Analysis of implementation data, teacher reflections, and iterative refinements revealed four interrelated design principles that proved both feasible and sustainable for embedding AI literacy in early childhood classrooms. These principles connect the children's observed learning outcomes with the pedagogical practices that supported them, illustrating how curriculum design can translate abstract AI concepts into developmentally meaningful experiences.

Embodied Play as Conceptual Grounding. Physical enactment and movement-based activities allowed children to externalize and test algorithmic thinking while engaging their bodies and imaginations.

Unplugged games such as *Program My Moves* or *Move Like a Robot* served as concrete anchors for understanding how AI systems follow human-generated rules. This embodied engagement bridged the abstract idea of algorithms with children's intuitive understanding of action sequences, promoting both comprehension and enjoyment.

Tangible Coding as a Procedural Bridge. Hands-on tools such as Bee-Bots provided visible and manipulable representations of algorithms and debugging processes. Sequencing, testing, and revising physical commands strengthened children's procedural understanding of how instructions produce predictable outcomes. These tangible experiences helped learners grasp core AI literacy concepts—control, causality, and iteration—within a play-based framework that supported agency and persistence.

Guided Dialogue as Ethical and Reflective Scaffolding. Teacher-facilitated discussions around fairness, data, and decision-making transformed everyday activities into opportunities for ethical reasoning. In activities such as *AI Stories and Me*, guided questioning (“Who told the robot what to say?” “Is it fair if the robot only listens to one person?”) encouraged children to recognize human agency in AI behavior and reflect on issues of equity and inclusion. Dialogue served as a social and cognitive scaffold for connecting technical understanding with moral awareness.

Teacher Co-Design as a Sustainability Anchor. Teachers' sustained participation in the co-design process ensured that the curriculum evolved in ways compatible with classroom realities. Their iterative input enhanced activity feasibility, alignment with standards, and adaptability across contexts. Co-design not only increased implementation fidelity but also fostered long-term ownership, confidence, and a culture of innovation among educators—key factors for sustaining early AI education beyond the study period.

Together, these design principles define a developmentally grounded model for introducing AI literacy in early learning contexts. By integrating embodied play, tangible coding, guided dialogue, and teacher co-design, PL-AI positioned both children and teachers as active participants in exploring how intelligent systems work, think, and relate to human values.

5. Discussion

The purpose of this study was to co-design, implement, and refine Play with AI (PL-AI)—a developmentally appropriate curriculum aimed at introducing foundational artificial intelligence (AI) concepts to pre-kindergarten and kindergarten learners through sequenced play, tangible coding, and guided dialogue with social robots. Using a design-based research (DBR) framework (Brown, 1992; Design-Based Research Collective, 2003), the researcher collaborated with early childhood teachers to iteratively develop, test, and refine activities that made abstract AI ideas accessible through embodied and interactive learning. Analyses of classroom interactions, teacher reflections, and formative documentation demonstrated that the PL-AI curriculum supported children's emerging AI-related reasoning while enhancing teachers' confidence, pedagogical agency, and curricular ownership.

This study contributes design-based evidence clarifying the pedagogical mechanisms through which AI literacy can be developmentally and ethically embedded in early childhood classrooms. Findings addressing [Research Question 1](#) indicated that teacher co-design shaped implementation quality, professional confidence, and curriculum ownership. Teachers' active participation in co-design and reflection shifted their roles from curriculum implementers to co-creators of pedagogical innovation. They adapted activities, refined materials, and aligned AI concepts with literacy, numeracy, and social-emotional learning goals—creating an integrated instructional approach rooted in familiar classroom routines. These outcomes are presented as indicators of teacher sensemaking and professional growth, rather than as measures of instructional effectiveness. Such findings align with prior research emphasizing that co-design strengthens adoption, contextual

fit, and sustainability of educational innovations (Penuel et al., 2007; Priestley et al., 2015; Voogt et al., 2015).

Findings related to [Research Question 2](#) indicated that the PL-AI curriculum supported conceptual, procedural, and dispositional dimensions of AI literacy. Conceptually, children began reasoning about AI systems as human-designed and instruction-following rather than autonomous. Procedurally, tangible coding experiences using Bee-Bots allowed children to plan, test, and debug sequences through play. Dispositionally, children displayed curiosity, persistence, and collaboration, often treating errors as opportunities for problem-solving. These observations are interpreted as formative indicators of emerging reasoning within play-based contexts, not as validated measures of learning outcomes. Ethical reflection also surfaced as a salient dimension, as guided discussions around fairness, bias, and human decision-making prompted children to consider how AI behavior reflects human choices and data. From a theoretical perspective, this study extends design-based research and early childhood pedagogy by articulating how embodied play, tangible coding, guided dialogue, and teacher co-design operate as interacting pedagogical mechanisms rather than isolated instructional strategies. These mechanisms align with frameworks that conceptualize learning as the integration of knowledge, practice, and disposition (Brennan & Resnick, 2012; NAEYC, 2021) and operationalize key AI4K12's Big Ideas—particularly Learning from Data and Societal Impact—within developmentally appropriate practice.

From an educational innovation perspective, PL-AI demonstrates how emerging AI education frameworks can be translated into early childhood settings without displacing core pedagogical values. Moving beyond framing AI as a stand-alone technology unit, PL-AI embeds AI concepts within existing classroom structures such as centers, storytelling, movement, and collaborative play. This integration addresses a persistent gap in the literature, where AI and computational thinking are often framed either as preparation for later computer science learning or as tool-based interventions divorced from early childhood pedagogy.

Through iterative refinement, four interrelated design principles emerged as feasible and sustainable for early AI education: embodied play as conceptual grounding, tangible coding as a procedural bridge, guided dialogue as ethical and reflective scaffolding, and teacher co-design as a sustainability anchor. Taken as an integrated model, these principles define a developmentally grounded model for introducing AI literacy that foregrounds human agency, ethical awareness, and collaborative inquiry. By positioning teachers and children as active participants in exploring how intelligent systems work and why they behave as they do, PL-AI contributes a replicable pedagogical approach to early AI education. Within this model, programming activities are not treated as vehicles for transmitting fixed claims about AI agency or animacy; instead, they function as dialogic spaces in which children are supported to negotiate meanings, examine evidence, and explore human values through guided discussion and shared inquiry (Bers, 2022).

Importantly, the methodological contribution of this study lies not in outcome measurement; instead, it demonstrates how DBR can be used to responsibly develop, refine, and study AI curricula in early childhood contexts. The findings illustrate how formative analytics, teacher reflection, and classroom enactment can inform design decisions while respecting the epistemological limits of small-scale, qualitative research. In this way, PL-AI advances both the practice and methodology of early AI education. PL-AI's most transferable contribution is the DBR-informed methodology and the four design principles (embodied play, tangible coding, guided dialogue, teacher co-design), which can be adapted to other early-childhood contexts and tools. PL-AI is intended to be portable at the level of design principles and teacher moves (e.g., inquiry prompts, guided dialogue routines), while remaining adaptable in its tools, activity sequencing, and local standards alignment. In contrast, specific activity sequences, robot platforms, and assessment tallies are context-dependent and should be treated as adaptable exemplars rather than prescribed requirements.

This study has several important limitations that should be

considered when interpreting the findings. First, as a design-based research project involving a small number of classrooms and participants, the study was not intended to produce statistically generalizable results or to establish causal relationships between the PL-AI curriculum and learning outcomes. Instead, findings are offered as context-sensitive, formative design evidence that informs feasibility, enactment, and pedagogical mechanisms.

Second, observational rubrics, participation tallies, and teacher self-reports were used as analytic supports for iterative curriculum refinement rather than as validated assessment instruments. Although these tools provided insight into patterns of engagement, reasoning, and teacher sensemaking, they should not be interpreted as standardized measures of AI literacy or instructional effectiveness. Relatedly, because societal understandings of AI and agency continue to evolve, PL-AI frames animacy and 'feelings' not as doctrinal claims but as inquiry topics; the analyses therefore focus on children's reasoning processes and evidence use rather than judging conceptual 'accuracy'.

Third, the implementation occurred within a limited set of early childhood contexts, all supported by participating teachers who volunteered for co-design. As a result, the curriculum's transferability to other settings—such as classrooms with different instructional constraints, levels of technological access, or teacher preparation—requires further investigation.

Finally, the study focused on short-term enactment and refinement cycles. Longitudinal research is needed to examine how early experiences with AI concepts influence children's later computational thinking, ethical reasoning, and digital citizenship, as well as how teachers' pedagogical practices evolve over time. Future studies employing complementary methodologies, including longitudinal and mixed-methods designs, could extend and strengthen the design principles identified in this work.

5.1. Educational implications

The findings of this study have important implications for early childhood education, teacher preparation, curriculum design, and educational policy. First, at the classroom level, the PL-AI curriculum demonstrates that AI literacy can be meaningfully integrated into early childhood settings without displacing core developmental priorities. By embedding AI concepts within familiar play-based structures—such as storytelling, movement, centers, and collaborative problem-solving—AI learning becomes a natural extension of early childhood pedagogy rather than a stand-alone technology unit. This approach supports children's engagement with foundational ideas such as algorithms, data, and fairness while maintaining alignment with developmentally appropriate practice (Bers, 2022; NAEYC, 2021).

Second, the findings highlight the importance of teacher co-design as a form of professional learning in emerging areas such as AI education. Rather than relying on tool-focused training, PL-AI illustrates how collaborative design, enactment, and reflection can support teachers' conceptual understanding, pedagogical confidence, and willingness to adapt AI-related instruction to local classroom contexts. Teacher education and professional development programs may therefore benefit from incorporating co-design models that position educators as active contributors to curriculum development, particularly when introducing complex and rapidly evolving domains like AI.

Third, from a curriculum design perspective, the PL-AI design principles offer practical guidance for adapting existing AI education frameworks to early childhood contexts. Embodied play, tangible coding, and guided dialogue function not as add-ons but as central pedagogical mechanisms that translate abstract AI ideas into child-meaningful experiences. Curriculum developers can draw on these principles to design AI learning experiences that foreground human agency, ethical reflection, and collaboration, rather than focusing narrowly on technical skills or tool use.

Finally, at the policy and systems level, this study underscores the

need for early inclusion of AI literacy within broader early STEM and digital literacy initiatives. Policies that support time, resources, and autonomy for teacher collaboration and iterative curriculum development are particularly critical for sustaining innovation in early childhood AI education. Investment in professional learning networks that connect researchers, curriculum designers, and early childhood educators may help ensure that AI literacy is introduced equitably and responsibly, beginning in the early years.

These implications position PL-AI not as a prescriptive curriculum, but as a flexible, developmentally appropriate framework that can inform future efforts to integrate AI literacy into existing early childhood curricula while honoring the field's longstanding commitment to play, inquiry, and social interaction.

5.2. Significance for future research

This study represents an early step in defining what developmentally appropriate AI education can look like in early childhood contexts. The PL-AI framework contributes to the growing body of research on AI literacy by demonstrating that computational, ethical, and socio-emotional learning can coexist in play-based environments. Future research should extend these findings through multi-site DBR studies that examine how co-design dynamics evolve across diverse educational systems, socioeconomic contexts, and teacher experience levels. Longitudinal investigations could trace how children's understanding of algorithms, data, and fairness develops over time and how early AI exposure influences later computational thinking and digital citizenship. Further, complementary methodologies, including mixed-methods and carefully designed quantitative approaches, could build on qualitative insights to examine how specific design principles (e.g., embodied play or guided dialogue) function across contexts, while remaining developmentally appropriate. Future work also could further formalize these concept categories using AI representations (e.g., an ontology or concept graph) to support systematic tagging of lesson features and classroom discourse, enabling a reusable knowledge base for early AI curriculum development and guideline work. These directions can refine theoretical models of early AI learning, inform teacher education frameworks, and guide equitable policy development as AI literacy becomes an essential dimension of 21st-century education.

Ethics statement

This curriculum design research was conducted in accordance with the ethical standards of the Institutional Review Board (IRB) at the University of Texas at Arlington. Ethical approval was obtained prior to implementation and data collection. All participating educators provided informed consent. Participation was voluntary, and participants were informed of their right to withdraw at any time without penalty. All data used to inform iterative curriculum refinement were de-identified to ensure confidentiality.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT (OpenAI) for language editing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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